

AFSHAN AZEEM

Data Science | Machine Learning | Analytics

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🐙 github.com/AfshaCodes

SUMMARY

Highly motivated Bioinformatics undergraduate with a robust technical background and a passion for leveraging computational technologies in genomic research. Demonstrated expertise in programming, data analysis, and cutting-edge genomic technologies with a proven track record of continuous learning and skill development. Seeking challenging internship opportunities in bioinformatics, data science, or computational biology to apply advanced technical skills, contribute to cutting-edge research, and prepare for an impactful career at the intersection of biology and technology.

WORK EXPERIENCE

AI engineer, Infinisai 07/2025 – ongoing

As a Junior AI Engineer, I work on building and deploying machine learning models for real-world applications. My role involves data preprocessing, model training, and utilizing tools such as Python, PyTorch, and TensorFlow. I also help improve model performance and support AI integration into products.

Research Intern, AIROMOB Consulting 01/2023 – 01/2025

Analyzed proprietary data models focusing on media monitoring, voice analytics, and sentiment/emotion analysis.

Developed understanding of industry applications in hospitality and data intelligence.

Mobile Development Intern, Grape Technologies 07/2022 – 09/2022

Gained hands-on experience in Android app development using Android Studio. Built simple mobile apps and learned core Java and XML development principles.

EDUCATION

BS Bioinformatics, COMSATS University Islamabad 03/2025

Relevant coursework: Data Analysis, Statistics, Machine Learning, Bioinformatics.

SKILLS

Languages

Python, R, SQL, C++, C#, Kotlin, HTML/CSS

Tools

Jupyter, GitHub, Power BI, SPSS, Flask

Soft Skills

Communication, Problem-Solving, Time Management, Teamwork

Libraries/Frameworks

Pandas, NumPy, Scikit-learn, Keras, Matplotlib, Seaborn

Bioinformatics

STRING, UniProt, BioGRID, TM-align, FPocket

PROJECTS

Gene-Drug Interaction Prediction in NSCLC

Built a computational pipeline using Random Forest, SVM, and XGBoost to predict gene-drug interactions. Integrated multi-feature analysis including sequence similarity, pocket similarity, and pathway data. Tools: Python, Scikit-learn, Pandas, Bioinformatics libraries. Achieved 92% model accuracy in validation tests using ensemble methods.

Deep Learning-Based Image Classification with Explainability

Developed and trained CNN, ResNet18, and MobileNetV2 models using Pytorch to classify biomedical images into "Normal" and "Stress" categories. And implemented Grad-CAM for visual explanation of model predictions to enhance interpretability.

Promoter Sequence Classifier Web App, *Python, PyTorch, Flask, Bioinformatics*

Built a deep learning-based tool to classify DNA promoter sequences (60bp) using CNN, BiLSTM, and Transformer models. Scraped 3800+ real sequences from EPD and generated balanced training data. Achieved >90% accuracy; applied Grad-CAM for interpretability. Deployed a Flask app that predicts promoters and fetches gene info from a local DB or EPD.

SMS Spam Detection

Built a binary classification model using Naive Bayes to detect spam messages. Improved accuracy to 96%.

Movie Recommendation System

Implemented a collaborative filtering-based recommender system using cosine similarity and pandas. Demonstrated results on a real-world movie dataset

CERTIFICATIONS

IBM Data Science Professional Certificate
(202)

Python for Genomic Data Science

Statistics for Genomic Data Science

Johns Hopkins (Coursera) - Genomic Data Science Series

Command Line Tools for Genomic Data Science

Introduction to Genomic Technologies

Algorithms for DNA Sequencing
Bioconductor for Genomic Data Science

Multomics - Scientific Writing & Data Analysis
(2022)